

AQUEDUCT_x

A generalized peer-to-peer logistics and finance layer
for smallholder farmers.

aqueductx.trade · open-source · MIT



Judgment doesn't scale down.

A third of the world's food is grown by smallholders. Between them and the markets that consume it stands a chain of intermediaries, because trade runs on judgment: someone must grade the goods, vouch for their origin, find the buyer, and carry the risk. That judgment is an operational cost a single smallholder lot cannot carry. And lenders do not lend against lots they cannot verify—**legibility comes before capital.**

\$2.5T

The global trade-finance gap—credit demanded worldwide that goes unmet.

ADB Trade Finance Gaps Survey, 2023 & Jan 2026

\$250

The minimum inspection fee per shipment—a smallholder lot cannot carry it.

Bureau Veritas Verigates PVoC fee schedule

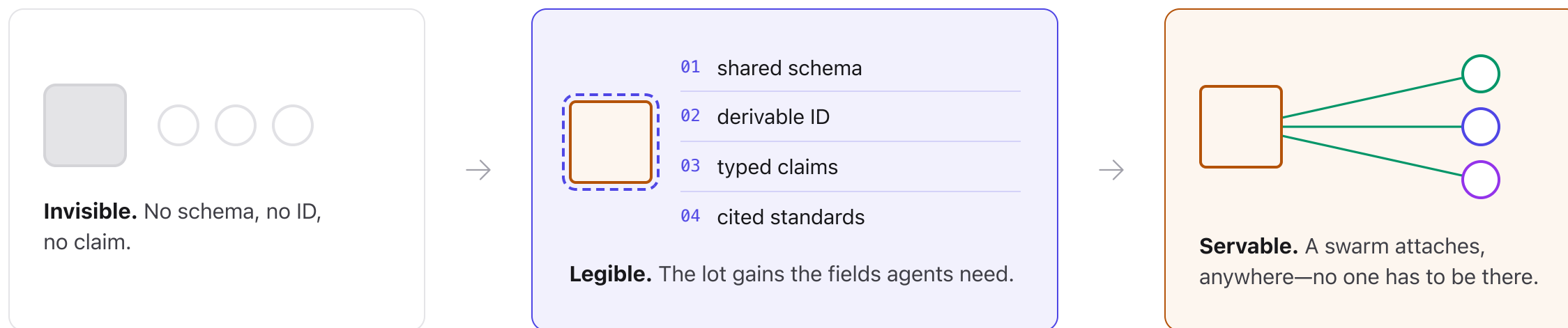
\$323B

Annual smallholder finance demand, against roughly \$95B supplied.

ISF Advisors Global Report, 2025

A legible lot is a servable lot.

A smallholder's lot fails three checks lenders require: it cannot afford professional verification, it must re-prove compliance to every new buyer, and nobody can confirm the harvest is not pledged twice. Make the lot legible—a shared schema, a derivable ID, typed claims—and agents run all three checks at almost no cost: one proof serves every standard, double-pledging becomes a registry lookup, and verification works at single-lot scale.

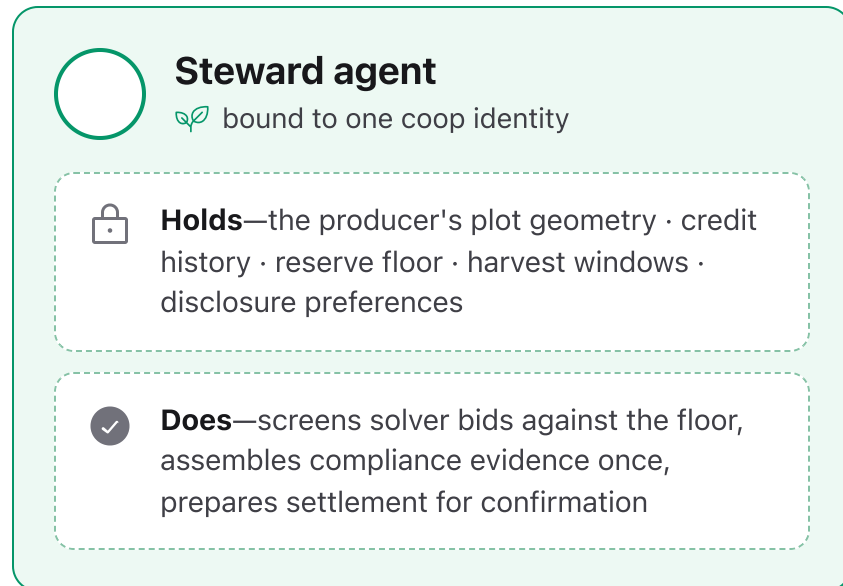


Sources: Bureau Veritas Verigates PVoC fee schedule (\$250 inspection floor) · MonetaGo duplicate-financing registry, live since 2018 · EU Deforestation Regulation (EUDR), in force.

The producer gets a seat, not a terminal.

Automated judgment placed in the farmer's hands—a steward agent whose loyalty is structurally to the producer.

THE SEAT



FIVE VERBS, ONE SURFACE

01 Post an intent

sell a lot, finance a planting, source an input.

02 Set the floor

the producer's reserve price; solvers bid against it, never under it.

03 Accept or decline a fill

matches arrive as named terms; the producer decides.

04 Confirm settle

human-confirmed by default—fully delegable to the steward by designation

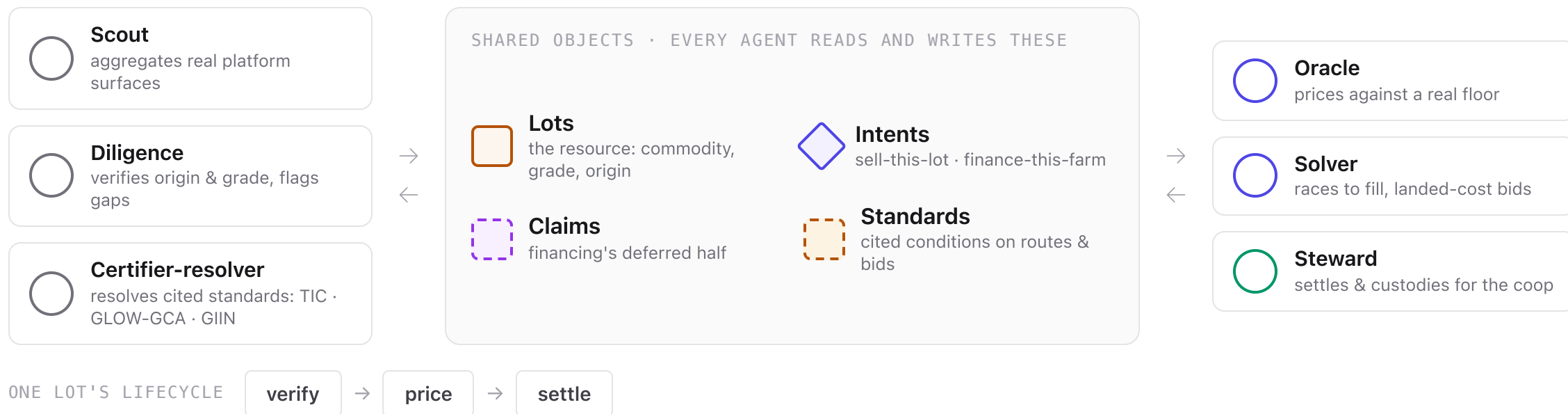
05 Set disclosure tiers

decides what evidence leaves the seat; plot geometry stays.

Stewards are extensible—producers can add negotiation strategies, alerts, or any automation that serves their interests. The verb set is a floor, not a ceiling.

One open layer, a swarm of agents.

Single-purpose agents act **concurrently** on a small set of shared objects. No agent owns the next; each reads and writes the same lots, intents, claims, and standards.



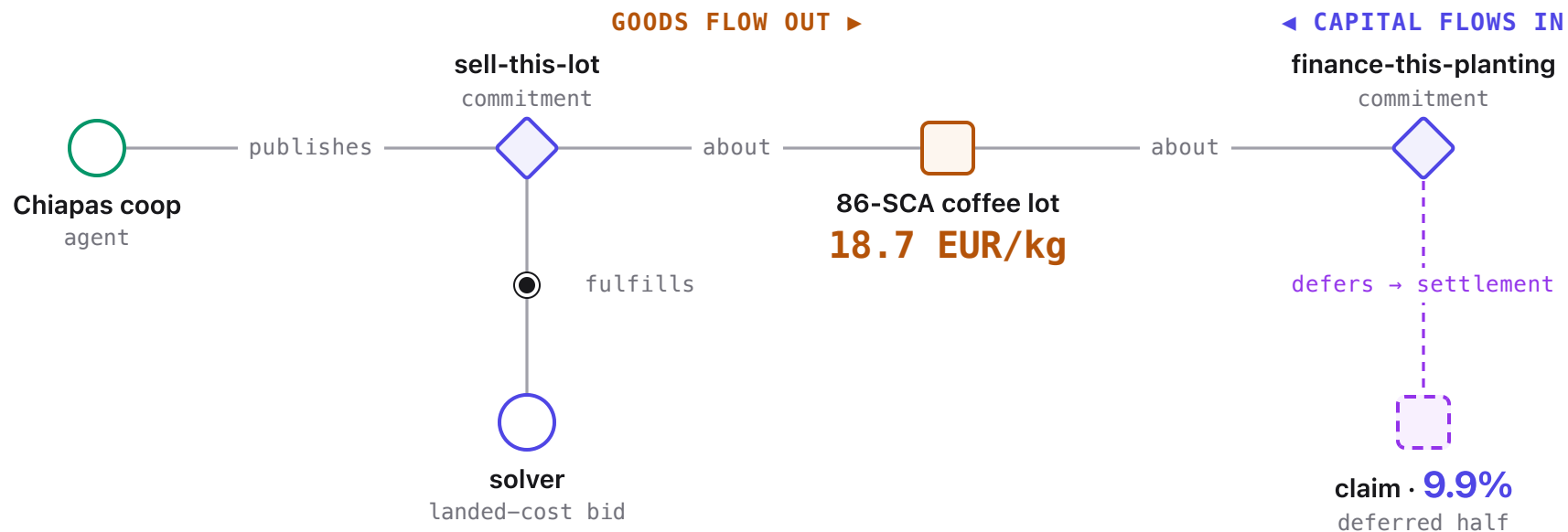
The data model is REA—the Claim is the point.

Selling a harvest and financing one are the same kind of exchange. AqueductX models the smallholder economy as **Resources, Events, and Agents**—what a farm sells and what it consumes to produce, in one schema. Writing the model down exposed the object every platform lacks: the **Claim**, a repayment owed later for capital advanced now.

REA CONCEPT	AQUEDUCTX TYPE	EXAMPLE
Resource	<code>AqueductAnyLot</code> + <code>inputResource</code>	Output —a lot the farm sells: coffee, weight, grade. Input —what the farm consumes, typed <code>inputResource: {resourceType}</code> : seedlings, home solar.
Agent	<code>AqueductActor</code> + hubs	cooperative, venue, solver, certifier, infrastructure
Commitment	<code>AqueductIntent</code>	sell-this-lot · finance-this-planting · finance-this-farm
Event	<code>AqueductEvent</code>	a fill—merged with real per-event source URLs
Claim	<code>claim: {principal, aprPct, term}</code>	A Claim is a repayment owed later for capital advanced now. The rate here is real: EthicHub lends to these cooperatives at a published 9.9% and has completed repayment cycles onchain. Source: EthicHub / Heifer facility documentation.

Coffee—the loop, running on real EthicHub reads.

A real Chiapas lot, read live from EthicHub. The coop publishes a sell-intent; a solver's fill fulfills it. Beside it, a finance-intent carries a Claim—capital now, repayment at the platform's published **9.9%**—the deferred half of the same exchange.

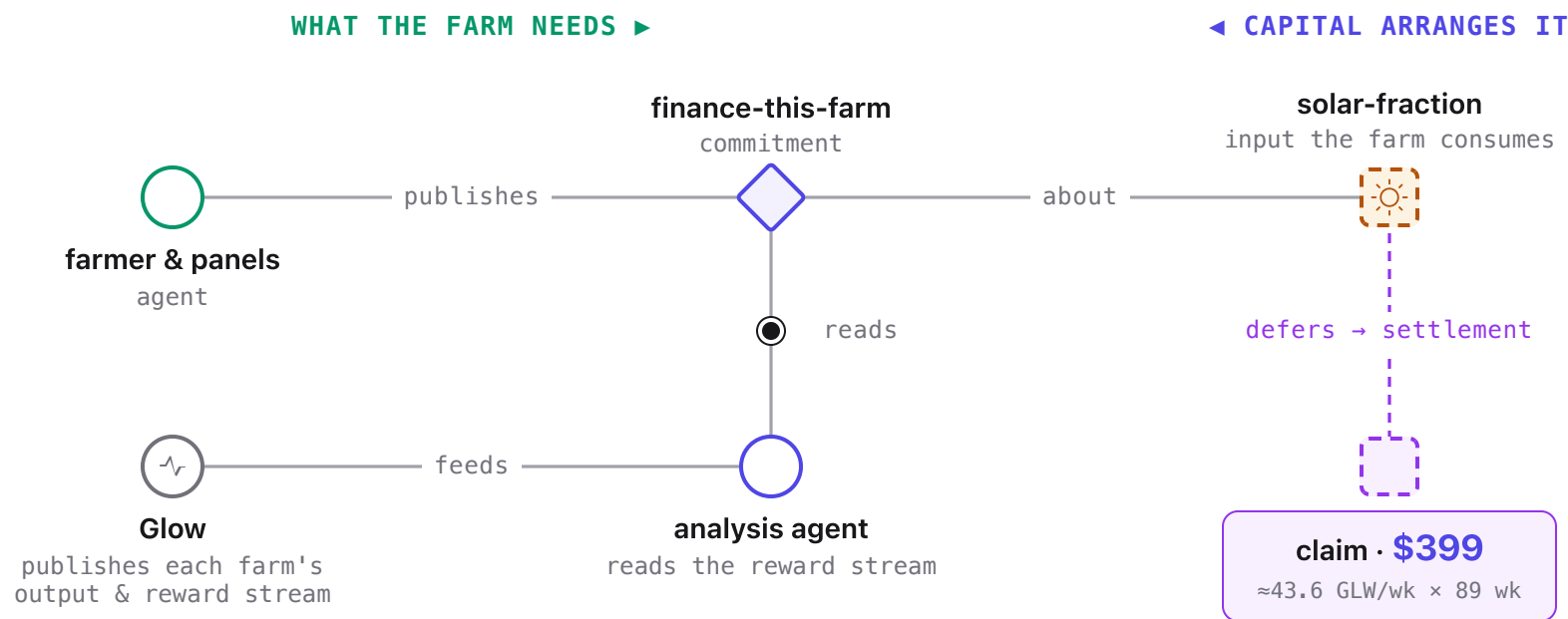


○ Agent □ Resource (lot) ◇ Commitment (intent) □ Claim (deferred) ● Event — typed relation - - - defers to settlement

Live read: greencoffee.ethichub.com · rate: EthicHub/Heifer facility

The same swarm that sells your coffee finances your panels.

A farmer needs energy to produce. An analysis agent reads the farm, finds the financeable input, and arranges the terms—the farmer sees the panel and the repayment.



124 audited farms read live · live GLW $\approx \$0.2825$ from the onchain pool—the agent reads a real stream, not a projection.

Two builders who ship, with institutional reach.



Patrick Rawson

Co-Founder & CEO

Full-stack builder across AI orchestration and blockchain mechanism design; ex-CEO of Curve Labs. Co-author (Taylor & Francis), *The Green Crypto Handbook: Blockchain for Sustainability Professionals*.

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Louise Borreani

Co-Founder

Researcher and strategist across AI, blockchain, and sustainability (MA Environmental Policy, Sciences Po Paris); authored the dMRV research underpinning Regen Atlas. Co-author, *The Green Crypto Handbook: Blockchain for Sustainability Professionals*.

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A TRACK RECORD THAT CONVERGED ON THIS

The Green Crypto Handbook

Rawson & Borreani, Taylor & Francis—the market thesis AqueductX operationalizes.



Regen Atlas

1st place, Infrastructure & Digital Rights, PL_Genesis (572 submissions)—the registry and map AqueductX is built on.



SlabClaw

Agent-legibility proven on a live market—the same rails now pointed at commodity lots.



AqueductX

The fusion: handbook thesis, atlas registry, and agent rails, live in one build.

Ecofrontiers SARL—advises central banks and international organizations · 🌐 ecofrontiers.xyz

\$50k buys four deliverables. **Public Good.**

Open-source MIT—agents, intent pipeline, and a reference solver to fork.

01 Harden three live verticals from existing partners on the Regen Atlas

EthicHub (coffee trade finance), Glow (solar finance), LandX (agri-commodity trade finance).

02 Wire the certification databases

Resolve every lot against real certifier registries—Fairtrade, Rainforest Alliance, TIC—verified at the source, not declared.

03 Sealed-bid solver races

Solvers bid without seeing each other's prices, so coops keep bargaining power.

04 Run a named-corridor pilot

First cooperative or exporter reading its own lots in through the ERP integration—the point where the registry stops being our data and starts being theirs.

aqueductx.trade

public MIT repo